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If m=1, then $\Delta T_b = K_b$;

Hence, the K_b is equal to the elevation in boiling point for 1 molal solution. K_b is calculated by the following expression

$$K_b = \frac{RT^2 M_{solvent}}{\Delta H_{vapourisation}}$$

Table 9.3 Molal boiling point elevation constant 'K_b' for some solvents

S. No.	Solvent	T_b° (K)	K_b (K kg mol^{-1})
1.	Water	373.15	0.52
2.	Ethanol	351.5	1.20
3.	Benzene	353.3	2.53
4.	Chloroform	334.4	3.63
5.	Ether	307.8	2.02
6.	Carbon tetrachloride	350.0	5.03
7.	Carbon disulphide	319.4	2.42
8.	Acetic acid	391.1	2.93
9.	Cyclohexane	353.74	2.79

Problem:

0.75 g of an unknown substance is dissolved in 200 g water. If the elevation of boiling point is 0.15 K and molal elevation constant is 7.5 K Kg mol^{-1} then, calculate the molar mass of unknown substance

$\Delta T_b = K_b m$

$= K_b \times W_2 \times 1000 / M_2 \times W_1$

$M_2 = K_b \times W_2 \times 1000 / \Delta T_b \times W_1$

$= 7.5 \times 0.75 \times 1000 / 0.15 \times 200$

$= 187.5$ g mol^{-1}

Evaluate Yourself

11) 2.56 g of Sulphur is dissolved in 100g of carbon disulphide. The solution boils at 319. 692 K . What is the molecular formula of Sulphur in solution The boiling point of CS_2 is 319. 450K. Given that K_b for CS_2 = 2.42 K Kg mol^{-1}

Depression in freezing point

Freezing point of a substance is another important physical property like boiling point. Freezing point is defined as "the temperature at which the solid and the liquid states of the substance have the same vapour pressure". At freezing point, the solid and liquid phases of the substance are in equilibrium. For example, the freezing point of water is 0 °C. At this temperature the ice and the water are in equilibrium. When a nonvolatile solute is added to water at its freezing point, the freezing point of the solution is lowered from 0 °C. The lowering of the freezing point of the solvent when a solute is added is called depression in freezing point (ΔT_f).

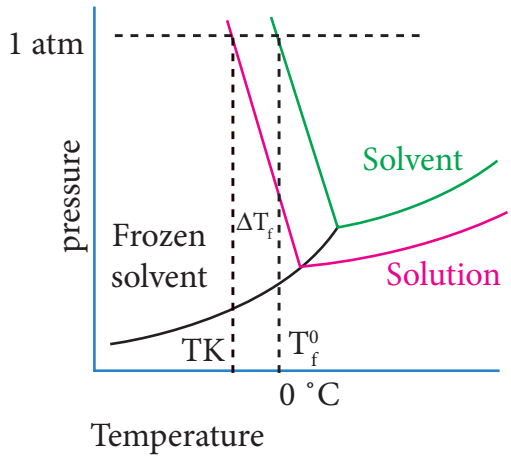


Fig 9.12 Depression in freezing point

From the above graph, we infer that the freezing point (T_f^0) is 0 °C as the